

ANKITA GAUTAM

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SUMMARY

Results-driven Data Scientist with over 3 years of hands-on experience in **data analysis, advanced statistical and predictive modeling, credit risk modeling, and machine learning** solutions. Adept at **data mining, data processing**, and translating complex data into actionable insights. Proficient in **Python, SQL, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, deep learning, NLP, and Generative AI**. Eager to contribute to high-impact roles that drive innovation, business growth, and strategic decision-making.

EXPERIENCE

Red Hat | Associate Data Scientist

FEB 2022 – present

Roles and Responsibilities:

- **Data Acquisition:** Collected, cleansed, and integrated data from multiple sources, ensuring high integrity, consistency, and reliability for accurate downstream analysis.
- **Exploratory Data Analysis:** Conducted in-depth **EDA** including both **Univariate** and **Multivariate analysis**, and created actionable visualizations using **Pandas, NumPy, Matplotlib, and Seaborn** to uncover trends, patterns, and data anomalies.
- **Machine Learning Modeling:** Designed and developed predictive models (**Linear Regression, Logistic Regression, Decision Tree, Random Forest, XGBoost, Neural Networks**), evaluated using metrics like **R2, RMSE, MSE, confusion matrix, AUC-ROC, F1-score, precision, and recall** for performance optimization.
- **Model Validation:** Applied rigorous validation techniques, including **cross-validation, backtesting, sensitivity analysis, hypothesis testing, and stress testing** to ensure model robustness, stability, and business relevance.
- **SQL & Data Transformation:** Utilized **SQL** for efficient data extraction, transformation, and aggregation from relational databases to support complex analyses.

Project: mode

- **Enhanced credit card risk model development** to address the economic shifts caused by the pandemic, improving its predictive power.
- Conducted comprehensive **EDA** to uncover key insights, resolving data issues like **outliers, missing values, and skewed features** through **advanced preprocessing techniques**, part of overall **model development** process.
- Redeveloped the model using cutting-edge **Random Forest**, fine-tuned with **hyperparameter optimization**, and validated rigorously through **sensitivity analysis, stress testing, and backtesting**.
- Delivered an **82%** accuracy rate, driving a **4%** business growth.

Project: Model Revalidation for Mortgage Loan Portfolios

- Revalidated the mortgage loan portfolio, assessing performance and risk profiles, and conducted **data cleaning, preprocessing, and in-depth Exploratory Data Analysis (EDA)** to identify key insights and patterns.
- Conducted **sensitivity analysis** of the model using both **Univariate and Multivariate testing** to assess the impact of changing inputs on the model's output.
- Performed **backtesting** for past 2 years and monitored the performance of the loans and updated the models as new data became available

Project: Positive Comment Ecosystems: Powered by Generative AI

- Designed a **GenAI** solution to filter and permit only constructive comments, fostering a positive online environment.
- Employed advanced techniques including **BERT-based sentence transformation, TF-IDF vectorization, word embeddings (Word2Vec), topic modeling (LDA), clustering (K-Means), and tokenization, stop word removal, lemmatization, stemming, and part-of-speech (POS) tagging** using the **NLTK** library.
- Built a **logistic regression model** in **Python**, supported by **cross-validation** and performance metrics such as **precision, recall, and F1-score** for robust comment classification.

SKILLS

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|---------------|-----------|------------------------|------------------------|-----------------------|
| • Python | • Pandas | • Matplotlib | • SQL | • Deep Learning |
| • Numpy | • Seaborn | • Machine Learning | • Scikit-learn | • Stability Analysis |
| • NLP | • GenAI | • Sensitivity Analysis | • Credit risk modeling | • Hypothesis testing |
| • Backtesting | • NLTK | • Impact Analysis | • Predictive modeling | • Statistical testing |

EDUCATION

B.E (Electronics and Telecommunication)

8 CGPA